

C977

Science Methods—Secondary Chemistry

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

In a unit planned during a busy grading week, students in a 30-seat chemistry room use data they collected to build a model explaining why some salts dissolve faster in warm water. Which part of this activity is the science and engineering practice?

- A. The warm water itself.
- B. Developing and using a model from their data.
- C. The idea that particle motion increases with temperature.
- D. Cause and effect.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A teacher wants students to use the crosscutting concept of energy and matter. Which prompt does that?

- A. Track where the energy goes when the hand warmer heats up, and where the atoms end up.
- B. Define the term exothermic and write it in your notebook.
- C. Any lesson about energy automatically uses the energy and matter lens.
- D. Plan an investigation to test the hand warmer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A chemistry teacher with a well-stocked lab and a new document camera is starting a unit on reaction rates. She has a striking demonstration of a reaction that speeds up when heated. In a phenomenon-driven 5E sequence, when should she introduce the formal term and model for collision theory?

- A. Only at the end of the unit, never during it.
- B. After students have observed the demonstration and explored variables that change the rate.
- C. Whenever the document camera is available to project notes.
- D. On the first day, before the demonstration, so students know what to look for.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Students in an honors section, most of whom earned A's last term, can balance the equation for burning methane. When asked to draw what happens to the particles, most draw the molecules simply disappearing. What is the best next teaching move?

- A. Assign more equation-balancing practice.
- B. Show a video of a gas-stove flame.
- C. Have students build or draw particle models showing atoms rearranging, and connect them to the equation.
- D. Move on, because honors students will pick it up later.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A student watches sugar vanish as it is stirred into water, then correctly explains, with a particle drawing, that the sugar molecules spread out among the water molecules. Which level has the student NOT yet shown?

- A. The crosscutting level.
- B. The particulate level.
- C. The symbolic level.
- D. The macroscopic level.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

During a lab in the school's newly renovated science wing, students react baking soda and vinegar in an open cup and find the mass went down. Many conclude that mass is destroyed in chemical reactions. The lab period is 50 minutes. What should the teacher do next?

- A. Have students repeat the reaction in a sealed bag and compare masses, then discuss where the gas went.
- B. Tell students the law of conservation of mass and move on.
- C. Shorten the next lab so there is time to lecture.
- D. Accept the conclusion because it matches their data.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A new chemistry teacher inherits a stockroom with a fresh coat of paint and a donated set of shelves. The previous teacher left the chemicals in one alphabetical row, so acetic acid sits beside ammonium nitrate. What is the correct way to reorganize?

- A. Store acids and oxidizers together because both are corrosive.
- B. Store everything on the new top shelves.
- C. Keep the single alphabetical row, because it makes chemicals easy to find.
- D. Separate chemicals into compatible families (acids, oxidizers, flammables, and so on) and alphabetize only within each family.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A chemical the teacher has never used before arrives for a lab. What should the teacher check first to learn its hazards, PPE and storage needs?

- A. The lab procedure from the textbook.
- B. Its Safety Data Sheet.
- C. A colleague who used it years ago.
- D. The label alone, which always contains everything needed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Students will titrate dilute hydrochloric acid with sodium hydroxide. One student wears thick prescription glasses and says they are shatterproof. The room has new lab tables. What eye protection should every student wear?

- A. Safety glasses with side shields.
- B. The student's prescription glasses, since they are shatterproof.
- C. None, because the acid is dilute.
- D. Indirectly vented chemical splash goggles, worn over prescription glasses if needed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

During a lab near the end of a long block, a student gets a splash of sodium hydroxide solution in one eye. The student's lab partner has the bottle label in hand and the class is behind schedule. What should happen first?

- A.** Flush with a weak acid to neutralize the base.
- B.** Read the SDS first to confirm the right treatment.
- C.** Flush for 1 to 2 minutes until the stinging stops.
- D.** Get the student to the eyewash and flush the eye with water for at least 15 minutes, then notify the office and parents.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

After 15 minutes at the eyewash, a student's eye still looks red. What is the next step?

- A.** Stop flushing, because 15 minutes always resolves any exposure.
- B.** Send the student back to finish the lab.
- C.** Get medical evaluation and complete the incident report.
- D.** Apply an eye ointment from the classroom kit.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A teacher with a generous technology grant and a class set of tablets wants students to see why increasing temperature raises gas pressure. Which use of technology best serves that goal?

- A.** A gas-particle simulation where students change temperature and watch particle speed and collisions change.
- B.** A slideshow app that students use to type definitions of pressure.
- C.** Buying more tablets so each student has two.
- D.** Replacing all gas-law labs with simulations for the rest of the year.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

In a unit ~~planned during a busy grading week~~, students in a ~~30-seat chemistry room~~ use data they collected to build a model explaining why some salts dissolve faster in warm water. Which part of this activity is the science and engineering practice?

Target word: which dimension is the practice **Decides:** Practices are what students do: developing and using models.

A	Camouflage	A lab material, not any of the three dimensions.
B	Correct	Building a model is one of the eight science and engineering practices.
C	Confusion set	That is a disciplinary core idea, the content.
D	Adjacent category	That is a crosscutting concept, a lens, not a practice.

2. Answer A Concept 1 • transfer

A teacher wants students to use the crosscutting concept of energy and matter. Which prompt does that?

Target word: a prompt that uses the lens **Decides:** The lens tracks flows of energy and matter through a system.

A	Correct	Tracing energy and matter through a system applies that crosscutting concept.
B	Adjacent category	Vocabulary recall is content, not the use of a lens.
C	Absolute language	The topic alone is not the lens; students must use it to reason.
D	Confusion set	Planning an investigation is a practice.

3. Answer B Concept 2

A chemistry teacher ~~with a well-stocked lab and a new document camera~~ is starting a unit on reaction rates. She has a striking demonstration of a reaction that speeds up when heated. In a phenomenon-driven 5E sequence, when should she introduce the formal term and model for collision theory?

Target word: when the formal explanation comes **Decides:** Explore comes before Explain.

A	Absolute language	Explain sits in the middle, followed by Elaborate and Evaluate.
B	Correct	Explanation builds on experience students have already had.
C	Camouflage	Equipment does not decide lesson sequence.
D	Inverted operation	Flips the order; the lab then only confirms what was told.

4. Answer C Concept 3

Students ~~in an honors section, most of whom earned A's last term~~, can balance the equation for burning methane. When asked to draw what happens to the particles, most draw the molecules simply disappearing. What is the best next teaching move?

Target word: the next move **Decides:** They have the symbolic level but not the particulate level.

A	Adjacent category	Strengthens the level they already have, not the missing one.
B	Confusion set	That is the macroscopic level; it does not show particles.
C	Correct	Bridges the symbolic and particulate levels.
D	Camouflage	Past grades do not show particle-level understanding.

5. Answer C Concept 3 • transfer

A student watches sugar vanish as it is stirred into water, then correctly explains, with a particle drawing, that the sugar molecules spread out among the water molecules. Which level has the student NOT yet shown?

Target word: missing level **Decides:** A particle drawing is the particulate level; no formula or notation is used.

A	Adjacent category	Not one of the three chemistry levels; mixes in a different framework.
B	Inverted operation	That is the level the drawing does show.
C	Correct	No formulas, equations or notation were used.
D	Confusion set	Watching the sugar vanish is the macroscopic level, which the student did show.

6. Answer A Concept 4

During a lab ~~in the school's newly renovated science wing~~, students react baking soda and vinegar in an open cup and find the mass went down. Many conclude that mass is destroyed in chemical reactions. ~~The lab period is 50 minutes.~~ What should the teacher do next?

Target word: teacher move for a misconception **Decides:** Confront the idea with contradicting evidence: a closed system.

A	Correct	Elicits, confronts and resolves the misconception with evidence.
B	Dropped element	Skips the confrontation step; the misconception tends to survive.
C	Camouflage	The period length does not decide the teaching move.
D	Inverted operation	Endorses the misconception instead of challenging it.

7. Answer D Concept 5

A new chemistry teacher inherits a stockroom ~~with a fresh coat of paint and a donated set of shelves~~. The previous teacher left the chemicals in one alphabetical row, so acetic acid sits beside ammonium nitrate. What is the correct way to reorganize?

Target word: storage method **Decides:** Storage is by compatible hazard family.

A	Confusion set	Many acids and oxidizers are incompatible and must be separated.
B	Camouflage	The new shelves do not decide compatibility; high shelves are also unsafe for liquids.
C	Adjacent category	Convenience is not a safety rule; it can put incompatible chemicals side by side.
D	Correct	Compatibility prevents dangerous reactions; alphabetizing inside a group is fine.

8. Answer B Concept 5 • transfer

A chemical the teacher has never used before arrives for a lab. What should the teacher check first to learn its hazards, PPE and storage needs?

Target word: first source to check **Decides:** The Safety Data Sheet carries hazards, PPE, storage and first aid.

A	Adjacent category	Procedures describe steps, not the full hazard profile.
B	Correct	The SDS lists hazards, protective equipment, storage and first-aid steps.
C	Confusion set	Experience helps but is not the authoritative hazard source.
D	Absolute language	Labels summarize; the SDS is the complete reference.

9. Answer D Concept 6

Students will titrate dilute hydrochloric acid with sodium hydroxide. One student wears thick prescription glasses and says they are shatterproof. ~~The room has new lab tables.~~ What eye protection should every student wear?

Target word: required eye protection **Decides:** Splashable acids and bases need sealed chemical splash goggles.

A	Adjacent category	They stop impact, not liquid splashes.
B	Camouflage	Answers the shatterproof detail; everyday glasses do not seal against liquids.
C	Absolute language	Dilute acids and bases can still injure eyes.
D	Correct	Goggles seal around the eyes against splashes.

10. Answer D Concept 7

During a lab ~~near the end of a long block~~, a student gets a splash of sodium hydroxide solution in one eye. ~~The student's lab partner has the bottle label in hand and the class is behind schedule.~~ What should happen first?

Target word: first action **Decides:** Immediate flushing for at least 15 minutes.

A	Inverted operation	Neutralizing in the eye adds heat and a second chemical; use water.
B	Camouflage	The label in hand tempts a delay; flushing comes first.
C	Dropped element	Leaves out the required duration; bases keep damaging tissue.
D	Correct	Immediate, prolonged flushing limits injury; reporting follows.

11. Answer C Concept 7 • transfer

After 15 minutes at the eyewash, a student's eye still looks red. What is the next step?

Target word: step after flushing **Decides:** Flushing is first aid; medical evaluation and reporting follow.

A	Absolute language	15 minutes is a minimum, not a guarantee.
B	Dropped element	Skips medical follow-up.
C	Correct	Exposures need follow-up care and documentation.
D	Adjacent category	Treatment is for medical staff, not the teacher.

12. Answer A Concept 8

A teacher ~~with a generous technology grant and a class set of tablets~~ wants students to see why increasing temperature raises gas pressure. Which use of technology best serves that goal?

Target word: tool that serves the learning goal **Decides:** The goal is to see particle behavior, which students cannot observe directly.

A	Correct	Makes the invisible particle level visible and lets students test the variable.
B	Adjacent category	Technology used for recall, not to reveal the mechanism.
C	Camouflage	Responds to the grant, not to the learning goal.
D	Absolute language	Simulations supplement safe hands-on labs; they should not replace them all.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	8	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The three dimensions of science	Science and engineering practices are what students do, crosscutting concepts are lenses that apply across all sciences, and disciplinary core ideas are the content; a three-dimensional lesson weaves all three together.	Calling a content topic a practice, or a lens such as energy and matter a core idea.	A classroom action verb (plan, model, argue) vs. a lens (patterns, cause and effect) vs. a content statement.
2	Phenomenon first: explore before explain	Anchor a unit in an observable phenomenon and let students explore it before formal explanation and vocabulary are given (Engage-Explore-Explain in the 5E model).	Opening with a vocabulary lecture and using the lab to confirm what was already told.	A question about lesson order or where the explanation belongs.
3	Macroscopic, particulate and symbolic levels	Chemistry understanding requires linking what is seen (macroscopic), what particles do (particulate), and how it is written (symbolic); teachers bridge through the particle level.	Assuming students who can balance an equation understand what the particles are doing.	Students succeed with symbols but fail to draw or explain particles.
4	Conceptual change for misconceptions	Durable misconceptions change when they are elicited, confronted with evidence that contradicts them, and resolved with a more useful model.	Telling students the right answer once and expecting the old idea to disappear.	A common wrong idea (for example, mass is lost when gas forms) and a choice of teacher move.
5	SDS and chemical storage by compatibility	Check the Safety Data Sheet before using a chemical, and store chemicals by compatible hazard family; alphabetical order is used only within a compatible group.	Storing the whole stockroom alphabetically.	A stockroom set-up or a new chemical arriving.
6	Eye protection matched to the hazard	Work with hazardous liquids, heat or splashes requires indirectly vented chemical splash goggles; ordinary safety glasses do not seal against splashes.	Treating any eyewear, including safety glasses or prescription glasses, as enough for chemical work.	Liquids that could splash, such as acids or bases.
7	First response to chemical exposure	For a chemical splash to the eyes or skin, flush immediately with water at the eyewash or shower for at least 15 minutes, then notify and get medical follow-up.	Neutralizing the chemical or waiting to identify it before flushing.	A splash has just happened; the question asks what comes first.

#	Concept	Rule	Trap	Tell
8	Purposeful technology integration	Choose technology because it serves the learning target, such as simulations that show particle behavior or reactions too dangerous to run; it supplements, not replaces, safe hands-on work.	Adding a tool because it is new, or replacing every safe lab with a simulation.	A tool choice justified by a learning need vs. by novelty.

Course scope (public description): Science Methods—Secondary Chemistry focuses on teaching methods specific to science for graduate students seeking an endorsement in secondary chemistry. Course content focuses on the design and teaching of standards-based lessons using the three dimensions of science (science and engineering practices, crosscutting concepts, and disciplinary core ideas) and the appropriate integration of technology into those lessons. Students in this course work within their content areas to evaluate, enhance, and plan appropriate science instruction. This course includes laboratory safety training and certification, which includes safe laboratory practices and procedures for science classrooms and the proper use of personal protective equipment. Previous coursework in curriculum, instruction, and assessment is a prerequisite for this course.